

Scalability & Future Plan

Date	08 July 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing Scalability and Future Strategy: and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority
1	Only supports local deployment	Cannot be accessed remotely by borrowers	High
2	SQLite database limits concurrent write performance	Not suitable for production-scale multi-user scenarios	High
3	No PDF export for generated letters	Users must manually copy letters for submission to lenders	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single/few users on local Flask/Uvicorn server	Deploy on cloud (Render/AWS/Vercel) with multiple Uvicorn workers behind Nginx
2	Data Storage	SQLite local file database	PostgreSQL on cloud (Supabase/AWS RDS) for enterprise-grade reliability
3	Performance	Local inference only	Redis caching for financial calculations, CDN for frontend assets
4	Security	JWT + bcrypt on local server	Add OAuth2 social login, 2FA, HTTPS/TLS encryption, rate limiting

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Cloud deployment (Render/Vercel) + Email verification + PDF letter export	2 months	Remote access for borrowers, professional letter downloads
Phase 3	Multi-language support (Hindi, Tamil, Telugu) + PostgreSQL migration + Mobile-responsive PWA	4 months	Wider user base, production-grade reliability, mobile accessibility
Phase 4	Mobile app (React Native) + Real-time notifications + Credit score integration	6 months	On-the-go debt management, proactive alerts, comprehensive financial picture